Theory Lunch, April 17

From Local to Global

Tushant Mittal

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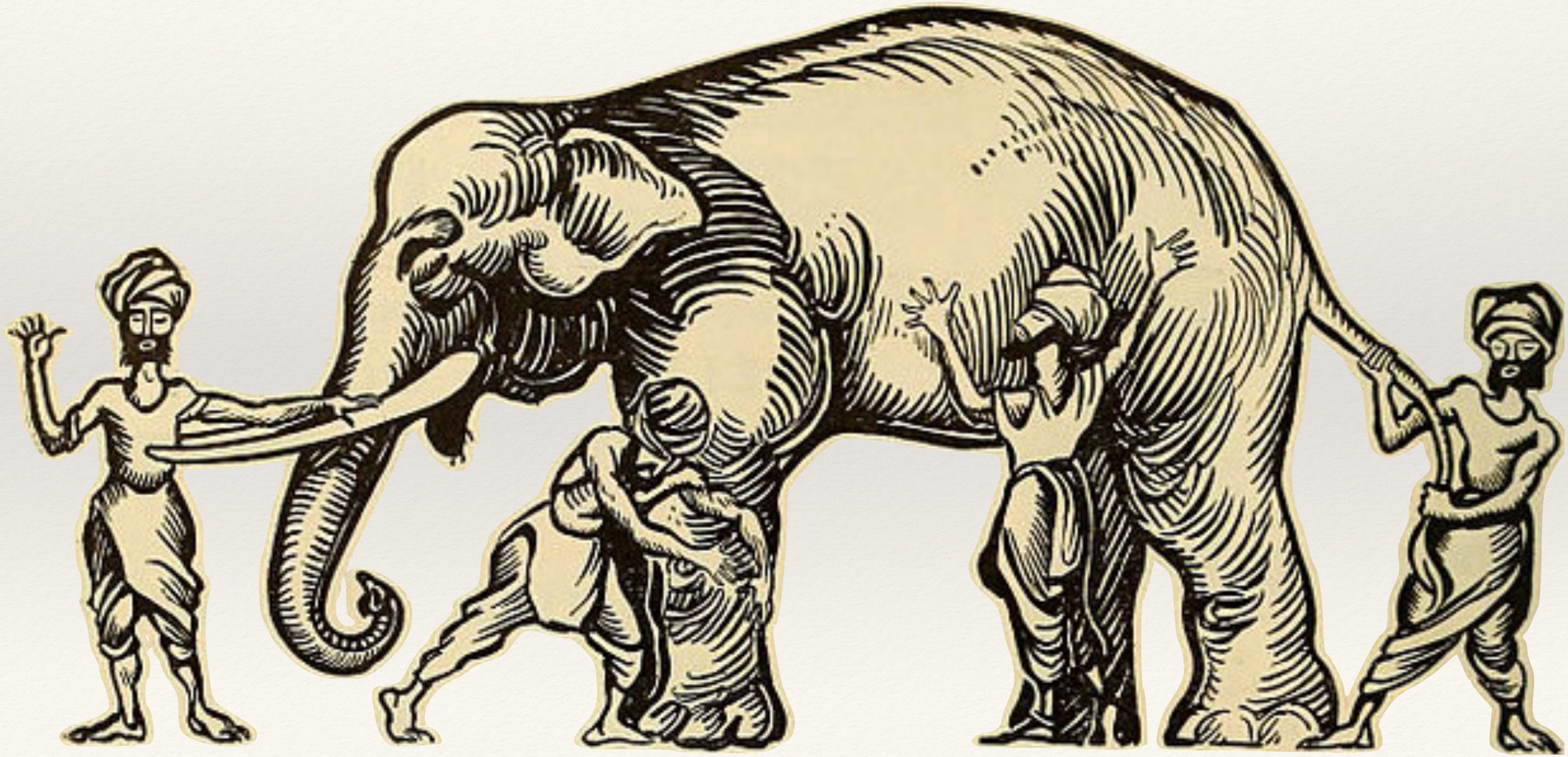
Sketches of an Elephant
A Topos Theory Compendium

VOLUME 2

PETER T. JOHNSTONE

When the blind men had each felt a part of the elephant, the king went to each of them and said to each: "Well, blind man, have you seen the elephant? Tell me, what sort of thing is an elephant?"

— *Udana 68-69: Parable of the Blind Men and the Elephant*



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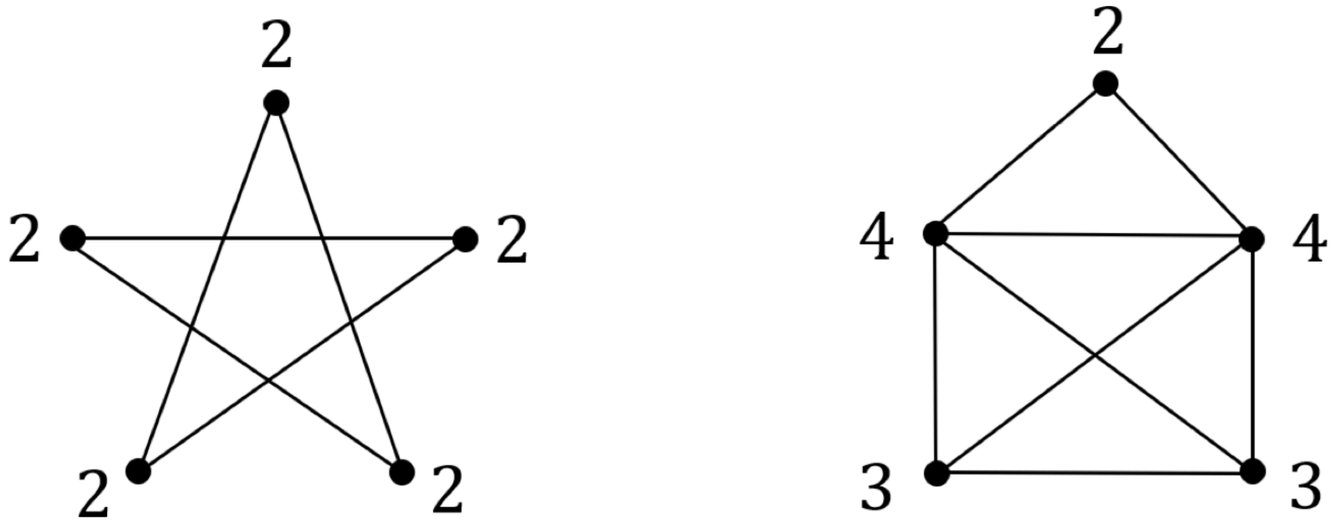
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- Hasse-Minkowski Theorem - *Let $f(x) = ax^2 + bx + c$. If f has a real root and root mod p^k for every prime p and positive integer k, then it has a rational root.*

A simpler example

Theorem: (Euler, 1735) *A connected graph has an Euler circuit if and only if every vertex has even degree.*

Recall that an Euler circuit is a path starting and ending at the same vertex which traverses each edge of the graph exactly once. According to Euler's Theorem, the left-hand graph in the diagram below has an Euler circuit because every vertex has degree 2, while the right-hand graph does not because the bottom two vertices have degree 3.



Euler's Theorem is an example of a local-global principle: the degrees of the vertices of a connected graph (a local property) tell you whether or not the graph has an Euler circuit (a global property).

More examples

From the blog, *Portrait of the Mathematician, Mathematics, statistics, philosophy, and way too much pop culture.*

https://harrisonbrown.wordpress.com/2010/01/01/whats-a-locally-determined-graph-property/

1. If every two vertices in a connected graph G have a unique common neighbor, then there's a vertex adjacent to all the others.
2. (Ore) If every pair of non-adjacent vertices in G has total degree at least n, then G is Hamiltonian.
3. If the subgraph spanned by every k vertices is bipartite, then G has chromatic number at most $O(n^{c/k})$ for some absolute constant c.
4. (Grotzsch) If G is planar and has no triangles, then G is 3-colorable.
5. (Dvorak-Kral-Thomas) If G is planar and every path between distinct triangles is longer than some absolute constant, then G is 3-colorable.
6. If no two vertices of G, both of degree greater than 2, are adjacent, then G is 3-colorable.
7. If G has roughly $n^2/4$ edges and about the same number of subgraphs isomorphic to C_4 as the Erdos-Renyi graph $G(n, 1/2)$, then all labeled graphs on k vertices occur about equally often as induced subgraphs of G, for $k << n$.

You get the idea!

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8. (Turan) If G is triangle-free, then $m \leq n^2/4$.
9. (Robertson-Seymour) The graph minors theorem.
10. If G contains no K_k subgraph and $n >> k$, then the complement of G must contain a K_k subgraph.
11. (Alon-Krivelevich) If a random induced subgraph of G of order $O(k\ ln\ k/\epsilon^2)$ is k-colorable, then with high probability you can delete ϵn^2 edges of G and obtain a k-colorable graph. (In particular, if every such subgraph is k-colorable, then the conclusion holds deterministically.)
12. (Conjecture, Erdos-Gyarfas) If G has minimum degree 3, then G contains a cycle with length a power of two.
13. (Conjecture, Hadwiger) If G contains no K_k minor, then G is (k-1)-colorable.
14. (Reconstruction conjecture) As long as n > 2, G is determined by the multiset of induced subgraphs on (n-1) vertices.

Clearly these statements are all over the place — some, like 6, are almost trivial to prove, while others take hundreds of pages or are still open! Furthermore, they run the gamut from extremal graph theory to topological problems to pure combinatorics.

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  - Homomorphism Testing, Agreement Testing, …

Graph Expansion

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- Expansion measures connectivity (global).
- $\lambda(X) = \max(\lambda_2(A), \lambda_n(A))$.
- Not actually a local property.
  - Global $\not\Rightarrow$Local

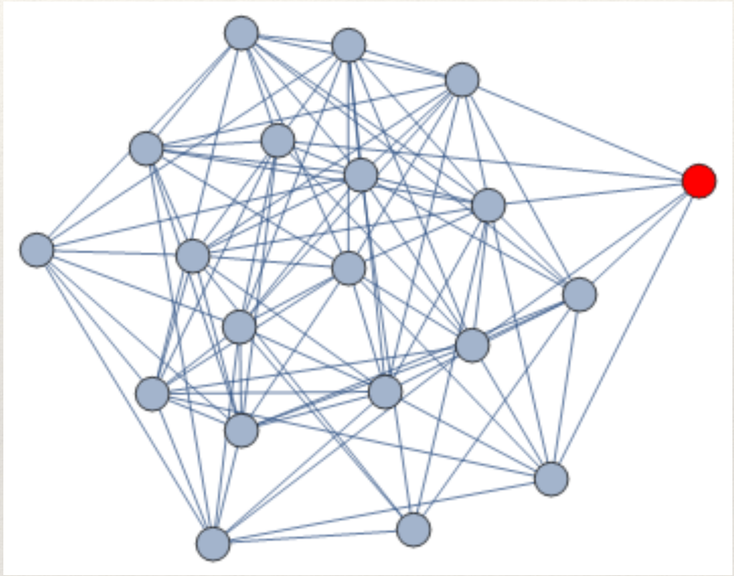


Image - ‘rhermans’ on Mathematica.SE

Expansion via Local Graphs

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- Trickle-Down Theorem (Oppenhein'17) — If X is connected and for every τ, X_τ is a λ-expander, then X is $\frac{\lambda}{1-(k-2)\lambda}$-expander.

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- Mihail—Vazirani'89 — Does this algorithm work for bases of any matroid?
  - Anari, Liu, Gharan, Vinzant'18 — Yes, via trickle-down!

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- Is the function f close ($\delta(\epsilon)$-close) to a true homomorphism?

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- Derandomization — Random sample from $(x, y) \sim S \subseteq G \times G$.

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  - Check if $g(A_k) = b_k$ for a random k.

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- Quantum Complexity, Entanglement Testing : $f : \mathscr{P}^n \to \mathbb{U}_d$ where the group is the Weyl-Heisenberg group or Pauli group modulo complex conjugation.

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- Something Else?